

AAE590FE Project

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1 Project Design

Students will propose an independent project related to their interests that involves the application or development of probabilistic estimation and tracking theory. Students will select one or more dimensions of complexity of their problem, to be identified in their project proposal and final report. These dimensions include:

- Filtering algorithm complexity - The student implements and applies a more sophisticated filtering algorithm, such as a Gaussian mixture or particle filter.
- Theoretical complexity - The student derives new mathematical results.
- Modeling complexity - The student derives models for a complex dynamical/sensing system and applies a filter for navigation or tracking.
- Analysis complexity - The student performs an extensive analysis covering multiple performance criteria, such as robustness to model mismatch, consistency, run-time performance, etc. This may also include comparison between two or more filters.
- Data complexity - The student applies a filtering technique to a real data set.
- Some other dimension of complexity, to be discussed with the instructor.

2 Project Proposal

Your project proposal should describe the problem you wish to solve, how you plan to solve it, what problems/challenges you might encounter, and how you will assess your outcomes. With that in mind, your project proposal should be at least one page long, consisting of the following sections:

1. Proposed problem description. Is this a tracking problem, a navigation problem, or something else? What is the state that you're trying to estimate? How will you model the dynamics, and are they linear or nonlinear? How will you model the measurements, and are they linear or nonlinear? Which "dimension(s) of complexity" does this problem fall under?
2. Proposed methodology. What filter(s) will you apply to solve the problem? You are encouraged to read ahead about algorithms not yet discussed in class. What components of the system do you plan to simulate, and what, if any, do you plan to extract from real data? For the simulated components, will you develop your own simulation, or use a third-party code? Will you use a higher-fidelity model in the simulation of the truth?
3. Anticipated challenges. Which aspects of the problem might pose the biggest challenges? Which aspects of the problem or methodology do you feel the least confident about or prepared to tackle? If part of your plan turns out to be too difficult or infeasible, how will you adjust your approach or rescope your project to complete something meaningful?

4. Expected results and proposed analysis. What will be the key result of your project? Will you look at one scenario, or multiple scenarios? How will you measure performance, and what plots and tables do you plan to produce? At a minimum, projects should present estimation error plots with 3σ bounds. Other aspects of performance include computation time, robustness, and consistency.

3 Final Report

Your final report will be typeset using L^AT_EX using an AIAA or IEEE template. For those who are new to L^AT_EX, overleaf.com is a very user-friendly platform for typesetting your report. Your report should contain at a minimum the following sections:

1. abstract
2. introduction
3. problem formulation
4. methodology
5. results and discussion
6. conclusions
7. appendices (for all code used in project)

4 Grading

Projects will be evaluated based on multiple criteria, including:

1. Report quality. Report is typeset in L^AT_EX and includes all relevant sections. All figures and tables are labeled and referenced. All axes and legends are labeled, include units, and are legible. All mathematical symbols are defined on first usage. All sources are appropriately cited.
2. Technical rigor. Equations are correct and written using consistent notation. All dynamics and sensor models are fully defined and include either a derivation or a citation to where the models were derived.
3. Problem complexity and creativity. Problem was non-trivial in at least one dimension of complexity. Your problem and approach is unique and does not resemble your classmate's project.
4. Quality of results. All results (negative or positive) are thoroughly explained and supported by theory, citations, and/or additional experiments. It's not acceptable to say something like "My results were poor because of numerical error"—you must somehow support your claim and propose next steps.

5 Key Due Dates

- 11:59 PM ET Dec 12, 2025 - Final Report Due
- 11:59 PM ET Oct 31, 2025 - Project Proposal Due

6 AI Use

Any use of AI not explicitly approved in writing by the instructor will be treated as academic dishonesty and will be handled accordingly. Note that receiving a zero in this project will lead to a failing course grade.

Ask your instructor beforehand if your proposed usage of AI is acceptable. An acceptable use may be, for example, generating a script that converts some real obscurely-formatted sensor data to a more usable format, like CSV. Not knowing how to use L^AT_EX is not an acceptable justification for having AI write your report.

7 Do not propose IMU-based navigation

IMU-based navigation filters are quite complex, and thus projects focused on these algorithms are typically too ambitious for this semester project. Navigation filters are commonly 15+ dimensional and estimate walking biases in the gyros and accelerometers. IMU errors are complex and comprised of white noise, randomly walking biases, start-up biases, temperature-dependent noises, scale factor errors, and orthogonality errors. Initialization of these filters is also complex and has its own name—“coarse alignment.”

These navigation filters are also different than the examples we’ve covered in class because, rather than use a dynamics model to predict motion, they simply use the IMU outputs. They also require some sort of external measurement to estimation position, such as GNSS (accelerometers only measure non-gravitational accelerations).